Diego Calderón

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Research Positions

since 10.2024	**Alexander von Humboldt/MPA Fellow**, Max Planck Institute for Astrophysics, Garching - Germany. Host: Prof. Dr. Selma de Mink.
03.2023 - 09.2024	**Postdoctoral Scientific Staff**, University of Hamburg, Hamburg - Germany. Adviser: Prof. Dr. Stephan Rosswog.
10.2019 - 02.2023	**Postdoctoral Researcher**, Charles University, Prague - Czech Republic. Adviser: Assoc. Prof. Dr. Ondřej Pejcha.

Education

Sep 2019	**Ph.D. in Astrophysics**, *summa cum laude*, Pontificia Universidad Católica de Chile. Thesis: *"The Role of Interacting Stellar Winds Feeding Sagittarius A*"*. Advisers: Prof. Dr. Jorge Cuadra & Dr. Marc Schartmann.
Mar 2015	**Master in Astrophysics**, *summa cum laude*, Pontificia Universidad Católica de Chile. Thesis: *"Thin-shell instabilities and clump formation due to colliding stellar winds in the Galactic Centre"*. Adviser: Prof. Dr. Jorge Cuadra.
Dec 2012	**Bachelor in Astronomy**, *summa cum laude*, Universidad de Concepción, Chile. Ranked 1 out of 35 enrolled at the Astronomy program in 2009.

Honors and Awards

Feb 2026	**Marie Skłodowska-Curie Fellowship**, European Commission, European Union.
Jul 2024	**Heraeus Fellowship**, Wilhelm and Else Heraeus Foundation, Germany.
Sep 2023	**Quantum Universe Postdoctoral Grant**, University of Hamburg, Germany.
Mar 2023	**Humboldt Research Fellowship**, Alexander von Humboldt Foundation, Germany.
2015-2019	**Doctoral Fellowship**, National Agency for Research and Development (ANID), Chile.
Mar 2015	**Doctoral Excellence Award**, Pontificia Universidad Católica de Chile.
2013-2014	**Master Fellowship**, National Agency for Research and Development (ANID), Chile.
2009-2012	**Undergraduate Scholarship**, Center for Excellence in Astrophysics and Associated Technologies (CATA).

Refereed Publications

Summary: 18 published (+2 submitted, +2 in prep.), 9 first-author papers (+2 in prep.).
NASA ADS (Mar 2026): 224 citations. Papers with students are shown in bold italic.

1. Abd El Dayem, K., Abuter, R., Aimar, N., P., A., Berdeu, A., Berger, J.-P., Bourdarot, G., Brandner, W., Burkert, A., **Calderón, D.**, Correia, C., Cuadra, J., Davies, R., Defrère, D., Delit, L., Drescher, A., Eisenhauer, F., Esteras Otal, L., Fabricius, M., … Zins, G. (2026). Discovery of a star sensitive to the spin of Sgr A*. *Submitted to Nature*.

2. Yilmaz, A., Lančová, D., Fürst, F., **Calderón, D.**, Serantes, S. R., Bozzo, E., Piano, G., Svoboda, J., Pittori, C., & Tavani, M. (2026). Discovery of a new candidate transient ultraluminous X-ray source in M51. *Submitted to A&A*.

3. **Calderón, D.**, Pejcha, O., Metzger, B. D., Duffell, P. C., & Rosswog, S. (2026). Quantifying the impact of relativistic precession on tidal disruption even light curves. *Astronomische Nachrichten*, e70092. https://doi.org/10.1002/asna.70092

4. Gillessen, S., Eisenhauer, F., Cuadra, J., Genzel, R., **Calderón**, **D.**, Joharle, S., Piran, T., Ribeiro, D. C., Russell, C. M. P., Sadun Bordoni, M., Burkert, A., Bourdarot, G., Drescher, A., Mang, F., Ott, T., Agapito, G., Agudo Berbel, A., Baruffolo, A., Bonaglia, M., … Xompero, M. (2026). The gas streamer G1-2-3 in the Galactic Center. *A&A*, *707*, A79. https://doi.org/10.1051/0004-6361/202555808

5. ***Kirilov, A.***, **Calderón**, **D.**, Pejcha, O., & Duffell, P. C. (2025). Two-dimensional Radiation-hydrodynamic Simulations of Luminous Red Novae. *ApJL*, *994*(2), L41. https://doi.org/10.3847/2041-8213/ae1ae7

6. ***Shaqil, W.***, **Calderón**, **D.**, Rosswog, S., Cuadra, J., Ciurlo, A., Morris, M. R., Campbell, R. D., & Ghez, A. M. (2025). Tidal phenomena in the Galactic Center: The Curious Case of X7. *A&A*, *700*, A134. https://doi.org/10.1051/0004-6361/202555191

7. Villanueva, V., Blaña, M., Bolatto, A. D., Rubio, M., Tarantino, E., Herrera-Camus, R., Burkert, A., Vaz, D., Read, J. I., Galaz, G., Muñoz, C., **Calderón**, **D.**, Behrendt, M., Carballo-Bello, J. A., Gray, E., & Fellhauer, M. (2025). The CHIMERA Survey: The first CO detection in Leo T, the lowest mass known galaxy still hosting cold molecular gas. *A&A*, *699*, L11. https://doi.org/10.1051/0004-6361/202555635

8. **Calderón**, **D.**, Cuadra, J., Russell, C. M. P., Burkert, A., Rosswog, S., & Balakrishnan, M. (2025). The formation and stability of a cold disc made out of stellar winds in the Galactic centre. *A&A*, *693*, A180. https://doi.org/10.1051/0004-6361/202452800

9. Blaña, M., Burkert, A., Fellhauer, M., **Calderón**, **D.**, Behrendt, M., & Schartmann, M. (2024). The Milky Way satellite galaxy Leo T: A perturbed cored dwarf. *A&A*, *692*, A183. https://doi.org/10.1051/0004-6361/202451649

10. Balakrishnan, M., Corrales, L., Markoff, S., Nowak, M., Haggard, D., Wang, Q. D., Neilsen, J., Russell, C. M. P., **Calderón**, **D.**, Cuadra, J., & Baganoff, F. (2024). Multistructured Accretion Flow of Sgr A*. I. Examination of a Radiatively Inefficient Accretion Flow Model. *ApJ*, *974*(1), 98. https://doi.org/10.3847/1538-4357/ad6c08

11. Balakrishnan, M., Russell, C. M. P., Corrales, L., **Calderón**, **D.**, Cuadra, J., Haggard, D., Markoff, S., Neilsen, J., Nowak, M., Wang, Q. D., & Baganoff, F. (2024). Multistructured Accretion Flow of Sgr A*. II. Signatures of a Cool Accretion Disk in Hydrodynamic Simulations of Stellar Winds. *ApJ*, *974*(1), 99. https://doi.org/10.3847/1538-4357/ad6866

12. **Calderón**, **D.**, Pejcha, O., Metzger, B. D., & Duffell, P. C. (2024). The effect of relativistic precession on light curves of tidal disruption events. *MNRAS*, *528*(2), 2568–2587. https://doi.org/10.1093/mnras/stae194

13. Pejcha, O., **Calderón**, **D.**, & Kurfürst, P. (2022). Supernovae in colliding-wind binaries: observational signatures in the first year. *MNRAS*, *510*(3), 3276–3290. https://doi.org/10.1093/mnras/stab3729

14. **Calderón**, **D.**, Pejcha, O., & Duffell, P. C. (2021). Moving-mesh radiation-hydrodynamic simulations of wind-reprocessed transients. *MNRAS*, *507*(1), 1092–1105. https://doi.org/10.1093/mnras/stab2219

15. Gómez, P. L., & Calderón, D. (2020). The Dynamics of the Wide-angle Tailed (WAT) Galaxy Cluster A562. *AJ*, *160*(4), 152. https://doi.org/10.3847/1538-3881/aba831

16. **Calderón**, **D.**, Cuadra, J., Schartmann, M., Burkert, A., Prieto, J., & Russell, C. M. P. (2020). Three-dimensional simulations of clump formation in stellar wind collisions. *MNRAS*, *493*(1), 447–467. https://doi.org/10.1093/mnras/staa090

17. **Calderón**, **D.**, Cuadra, J., Schartmann, M., Burkert, A., & Russell, C. M. P. (2020). Stellar Winds Pump the Heart of the Milky Way. *ApJL*, *888*(1), L2. https://doi.org/10.3847/2041-8213/ab5e81

18. **Calderón**, **D.**, Cuadra, J., Schartmann, M., Burkert, A., Plewa, P., Eisenhauer, F., & Habibi, M. (2018). The Galactic Centre source G2 was unlikely born in any of the known massive binaries. *MNRAS*, *478*(3), 3494–3505. https://doi.org/10.1093/mnras/sty1330

19 **Calderón, D.**, Bauer, F. E., Veilleux, S., Graciá-Carpio, J., Sturm, E., Lira, P., Schulze, S., & Kim, S. (2016). Searching for molecular outflows in hyperluminous infrared galaxies. *MNRAS*, *460*(3), 3052–3062. https://doi.org/10.1093/mnras/stw1210

20 **Calderón, D.**, Ballone, A., Cuadra, J., Schartmann, M., Burkert, A., & Gillessen, S. (2016). Clump formation through colliding stellar winds in the Galactic Centre. *MNRAS*, *455*(4), 4388–4398. https://doi.org/10.1093/mnras/stv2644

ADS library link

Service & Skills

Coding	C, FORTRAN, Python, SHELL scripts.
Languages	Czech (B1), English (C1), German (A2), Spanish (mother tongue).
Professional	SOC member at XIX Chilean Astronomical Society Annual Meeting (Nov 2024).
	SOC member at 2nd Workshop of Astronomy Students (March 2017).
Referee	Astronomy & Astrophysics (1), Astrophysical Journal (3), Czech Science Agency (3), Israel Science Foundation.
Representation	Postdoctoral Representative 2025 at Max Planck Institute for Astrophysics.
Supervision	Timea Unold - Master student at Technical University of Munich - since Oct 2025.
	Anthony Kirilov - Ph.D. student at Charles University - expected Aug 2026.
	Wasif Shaqil - Master student at University of Hamburg - Apr 2025. Next position Ph.D. student at Liverpool John Moores University, United Kingdom.
Thesis Committee	Michal Jireš - Bachelor in Physics at Charles University - Feb 2024.
	Yelyzaveta Pulnova - Master in Physics at Charles University - Sep 2021.

Teaching Experience

2024-2025	Teaching assistant at Ludwig Maximilian University of Munich in graduate course: *"Astrophysics Seminar"* (Fall 2024, Spring 2025, Fall 2025).
2023-2024	Teaching assistant at University of Hamburg in undergraduate course: *"Introduction to Astronomy"* (Spring 2023); and graduate course: *"Astrophysical Hydrodynamics"* (Fall 2023).
2019 - 2023	Co-organiser of Science Coffee (also as a class), Charles University.
Oct 2021	Special lecture: *"Introduction to Radiation Hydrodynamics"*, Charles University.
2017	Science Coffee (in charge once per week), Pontificia Universidad Católica de Chile.
2015-2017	Teaching Assistant at Pontificia Universidad Católica de Chile in undergraduate courses: *"Extragalactic Astrophysics"* (Fall 2017), *"Introduction to Physics"* (Spring 2015), *"Introduction to Astronomy"* (Fall 2015); and graduate courses: *"Galactic Centre Astrophysics"* (Spring 2016), *"Radiative Processes in Astrophysics"* (Fall 2016).
2011-2012	Teaching Assistant at Universidad de Concepción in undergraduate courses: *"General Astrophysics"* (Fall 2012), and *"Electromagnetic Theory"* (Spring 2011).

Funding and Grants

Feb 2026	**Marie Skłodowska-Curie Fellowship** (∼200 k€), European Commission, Europe.
2025-2026	**Cycle 4 JWST Theory grant #8655** (PI: C. Russell) as co-Investigator, USA.
2025-2028	**FONDECYT Regular 2025** (PI: J. Cuadra), as co-Investigator, ANID[1], Chile.
Jul 2024	**Heraeus Fellowship** (∼5 k€), Wilhelm and Else Heraeus Foundation, Germany.
Sep 2023	**Quantum Universe Postdoctoral Grant** (10 k€), University of Hamburg, Germany.

[1]ANID: National Agency for Research and Development

Funding and Grants (continued)

- Mar 2023 — **Humboldt Fellowship** (∼100 k€), Alexander von Humboldt Foundation, Germany.
- Jan 2023 — **2,210,400 CPU hours awarded**, IT4Innovations Supercomputing Center, Czechia.
- 2022-2023 — **Cycle 24 Chandra Theory grant #24200412** (PI: C. Russell) as co-Investigator, USA.
- Mar 2022 — **936,000 CPU hours**, IT4Innovations National Supercomputing Center, Czechia.
- 2021-2022 — **Cycle 23 Chandra Theory grant #23610258** (PI: C. Russell) as co-Investigator, USA.
- 2015-2019 — **Doctoral Fellowship** (∼45 k€), ANID, Chile.
  - Plus internship abroad for 10 months and travel budget for 2 years (total: ∼20 k€).
- Mar 2015 — **Doctoral Excellence Award** (∼10 k€), Pontificia Universidad Católica de Chile.
- 2014,2016,2018 — **Adelina Gutiérrez Travel Grant**, Chilean Astronomical Society (SOCHIAS).
- 2013-2014 — **Master Fellowship** (∼15 k€), ANID, Chile.
- 2009-2012 — **Undergraduate Scholarship** (∼8 k€), Center for Excellence in Astrophysics and Associated Technologies (CATA).

Internships, Extended Workshops, and Summer Schools

- Aug 2026 — MIAPbP program: *Bridging Gaps in High-Mass Star Research: Physical Parameters, Formation and Evolution*, Garching – Germany.
- Jul 2024 — Special Meeting: 73rd Lindau Nobel Laureate Meeting, Lindau – Germany.
- Jul 2023 — School: Kavli Summer Program in Astrophysics, Garching – Germany.
- Jul 2022 — Workshop: *"Plasma Under Strong Gravity"*, Aspen Center for Physics, Aspen – USA.
- Feb - Nov 2018 — **Visiting Ph.D. Student**, Max Planck Institute for Extraterrestrial Physics - Germany.
  Hosts: Prof. Dr. Andreas Burkert & Dr. Marc Schartmann.
- Aug 2017 — School: *"Astrophysical Plasmas"*, Niels Bohr Institute, Copenhague – Denmark.
- Apr 2015 — School *"Scientific Writing for Young Astronomers"*, EDP Sciences, Puerto Varas – Chile.
- Jun 2014 — School: *"Galaxies: Near and Far, Young and Old"*, Vatican Observatory – Vatican City.
- Jan - Mar 2013 — **Research Intern**, Cerro Tololo Inter-American Observatory - La Serena, Chile.
  Adviser: Dr. Percy Gómez.

Invited Conference Talks

- May 2025 — 19th International Max Planck Research School Symposium - Germany.
- Jul 2024 — 17th Marcel Grossmann Meeting - Pescara, Italy.
- Feb 2024 — Quantum Universe Postdoctoral Grant Prize talk at DESY - Hamburg, Germany.
- Apr 2023 — Galactic Center Workshop - Granada, Spain.

Conference Oral Contributions

- May 2026 — IAU Symposium 405: *"Traversing the Galactic Center in Space and Time"* - Brno, Czechia.
- Nov 2025 — RAGTIME27 Workshop - Opava, Czech Republic.
- Jun 2025 — *"X-ray QPEs and Repeating Nuclear Transients"* Workshop - Madrid, Spain.
- Sep 2024 — *"X-ray transients, theory and observational perspectives"* Conference - Warsaw, Poland.
- *"Tidal Disruption Events and Nuclear Transients"* Conference - Heraklion, Greece.
- Apr 2024 — RAMSES User Meeting 2024 - New York, USA.
- Dec 2023 — *"Fundamental Physics at the Galactic Centre"* Workshop - Porto, Portugal.
- Jul 2023 — *"Black-hole accretion discs 50 years on: current fortunes and new challenges"* and *"Supermassive Black Hole Transients and Their Host Galaxies"* Special Sessions at EAS - Kraków, Poland.

Conference Oral Contributions (continued)

Apr 2023	*"Quantum Universe Day"* at DESY - Hamburg, Germany.
Nov 2022	6th Users' Conference of IT4Innovations - Ostrava, Czechia.
Oct 2022	RAGTIME24 Workshop - Opava, Czechia.
Sep 2022	31st Texas Symposium on Relativistic Astrophysics - Prague, Czechia
Nov 2021	*"The Predictive Power of Computational Astrophysics as a Discovery Tool"* Symposium - Online.
Sep 2021	RAGTIME23 Workshop - Opava, Czechia.
Feb 2021	43rd COSPAR Assembly - Online Symposium.
Jan 2021	XVI Chilean Astronomical Society (SOCHIAS) Annual Meeting - Online.
Jul 2020	European Astronomical Society (EAS) Annual Meeting - Online.
Oct 2019	*"Common-Envelope Evolution and Post-Common-Envelope Systems"* Workshop - London, UK.
Sep 2018	RAMSES User Meeting 2018 - Lyon, France.
Jul 2018	42nd COSPAR Assembly - Pasadena, USA.
Nov 2016	XV Latin American Regional IAU Meeting (LARIM) - Cartagena, Colombia.
Mar 2016	XIII Chilean Astronomical Society (SOCHIAS) Annual Meeting - Antofagasta, Chile.
Feb 2016	*"Dynamics and Accretion at the Galactic Center"* Workshop - Aspen, USA.
Mar 2015	XII Chilean Astronomical Society (SOCHIAS) Annual Meeting - Puerto Varas, Chile.
Jun 2014	European Week of Astronomy and Space Sciences (EWASS) - Geneva, Switzerland.

Invited Talks at Research Institutions

Mar 2026	Astronomy seminar at Universidad Federico Santa María - Valparaíso, Chile.
Aug 2025	Seminar at Astronomical Institute, Czech Academy of Sciences - Prague, Czechia.
Mar 2025	Astronomy seminar at Universidad de Santiago de Chile - Santiago, Chile.
	Astronomy seminar at Universidad de Concepción - Concepción, Chile.
Feb 2025	Institute seminar at Max Planck Institute for Astrophysics - Garching, Germany.
Jan 2025	Seminar at Departamento de Astronomía, Universidad de Chile - Santiago, Chile.
Apr 2024	Astrophysics seminar at University of Delaware - Newark, USA.
Jan 2024	Astrophysics seminar at Ludwig Maximilian University - Munich, Germany.
Oct 2023	Astronomy seminar at Universidad de Valparaíso - Valparaíso, Chile.
	Astronomy seminar at Universidad Federico Santa María - Valparaíso, Chile.
Sep 2023	Astronomy colloquium at Universidad de Concepción - Concepción, Chile.
Jul 2023	Astrophysics seminar at Ludwig Maximilian University - Munich, Germany.
Jun 2023	Observatory Colloquium at Hamburg Observatory - Hamburg, Germany.
Oct 2021	Relativistic Seminar at Silesian University - Opava, Czechia.
Sep 2021	Astronomy seminar at Masaryk University - Brno, Czech Republic.
Mar 2021	Stellar seminar at Czech Academy of Sciences - Online.
Feb 2021	Relativistic seminar at Charles University - Online.
Apr 2020	ELI Beamlines Scientific Seminar - Dolní Břežany, Czech Republic.
Feb 2020	Numerical Mathematics Seminar at Charles University - Prague, Czech Republic.
Oct 2019	Japan Aerospace Exploration Agency (JAXA) High-energy Seminar - Sagamihara, Japan.
Aug 2019	European Southern Observatory (ESO) Seminar - Santiago, Chile.
Oct 2018	G2G on WR140 Workshop at Royal Observatory of Edinburgh - Edinburgh, Scotland.
Sep 2018	Centre for mathematical Plasma-Astrophysics seminar at KU Leuven - Leuven, Belgium.

Invited Talks at Research Institutions (continued)

- Jul 2017 — High-energy Astrophysics Seminar at Max Planck Institute for Astrophysics - Germany.
- Feb 2017 — Infrared group Tea meeting at Max Planck Institute for Extraterrestrial Physics - Germany.
- Jun 2016 — Astronomy seminar at Universidad de Concepción - Concepción, Chile.

Conference Poster Contributions

- Nov 2022 — *"The Impact of Binaries on Stellar Evolution"* Workshop - Garching, Germany.
- Jul 2020 — European Astronomical Society (EAS) Annual Meeting - Leiden, Netherlands.
- Oct 2019 — *"New Horizons in Galactic Center Astronomy and Beyond"* Workshop - Yokohama, Japan.
- Jun 2017 — European Week of Astronomy and Space Sciences (EWASS) - Prague, Czech Republic.
- Oct 2016 — XV Latin American Regional IAU Meeting (LARIM) - Cartagena, Colombia.
- Jul 2016 — *"The Multi-Messenger Astrophysics of the Galactic Centre"* Symposium - Cairns, Australia.
- Jul 2015 — International Workshop on Wolf-Rayet stars - Potsdam, Germany.
- Mar 2015 — *"Unveiling the AGN-Galaxy Evolution Connection"* Workshop - Puerto Varas, Chile.
- Jul 2014 — European Week of Astronomy and Space Sciences (EWASS) - Geneva, Switzerland.
- Jan 2014 — 223rd American Astronomical Society Meeting - Baltimore, USA.
- Nov 2013 — XIV Latin American Regional IAU Meeting (LARIM) - Florianopolis, Brazil.

Outreach Efforts

- May 2025 — *"Lunch with an Astrophysicist"* at Max Planck Institute for Astrophysics, Germany.
- Jul 2024 — Galactic Centre Virtual Reality at *"Sternstunden Festival"*, Hamburg Observatory.
- Jun 2024 — Galactic Centre Virtual Reality at *"Quantum Universe Day"* at DESY in Hamburg.
- 2023-2024 — Editor of the Hamburg Observatory Newsletter.
- Jul 2023 — *"Ask an Astronomer"* and Tour Guide at *"Sternstunden Festival"*, Hamburg Observatory.
- 2020 — Interview at online newspaper *"El Contraste"* during Solar Eclipse 2020, Chile.
- 2014-2016 — Quantum Physics course for high-school teachers, Pontificia Universidad Católica de Chile.
- 2015 — Special lectures for students during the *"Day of Astronomy"* in Curicó, Chile.
- 2013-2014 — Physics courses for high-school students at Pontificia Universidad Católica de Chile.
- 2011-2012 — Guide of public astronomical observations at Universidad de Concepción, Chile.

References

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